

EE446 LAB 7 REPORT

Serial monitor results before fine tuning for digits 0-9

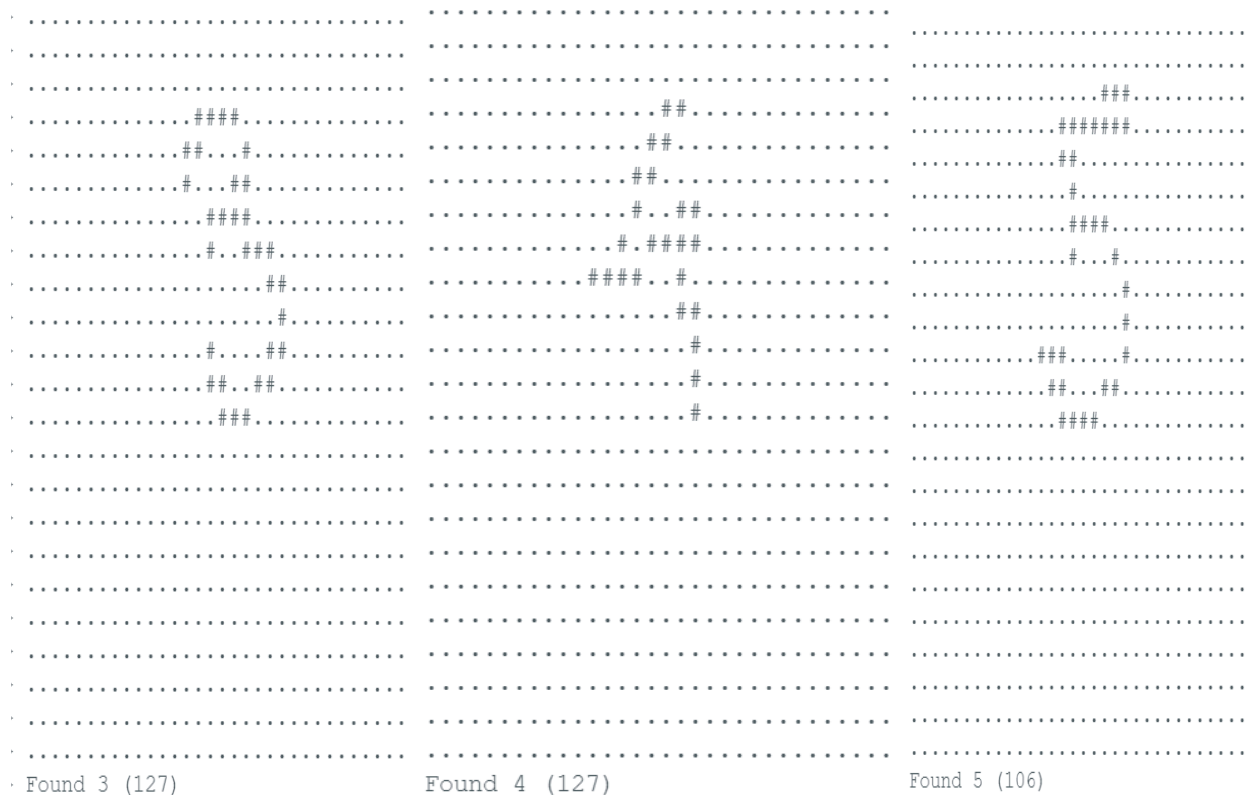
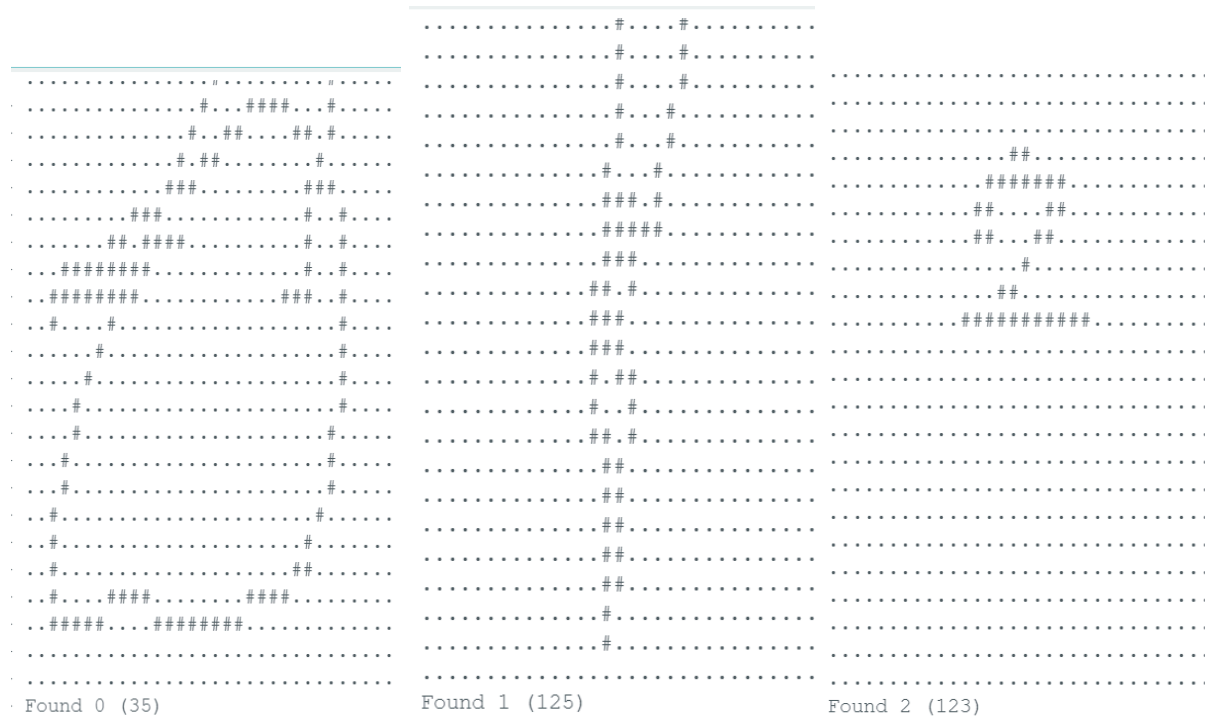
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Serial monitor results with fine tuning for digits 0-9



Before fine-tuning, the pretrained CNN model performed poorly on my collected gesture dataset, achieving only 8% accuracy. Most classes had very low precision, recall, and F1-scores, and the confusion matrix showed that the model frequently confused different digits. This happened because the pretrained model was not adapted to my personal gesture style and motion patterns while drawing digits in the air.

After fine-tuning the model using my own collected data, the performance improved significantly to 54% accuracy. Several classes such as digits 1, 2, 3, 6, and 9 showed much higher recall and precision scores. The confusion matrix after fine-tuning had more values concentrated along the diagonal, indicating that the model was making more correct predictions. Additionally, the model gave higher confidence scores on the serial monitor for the fine-tuned model.

2) Which model performed better and why?

The fine-tuned model performed much better than the original pretrained model. Fine-tuning allowed CNN to learn the unique characteristics of the motion sensor data collected from my gesture patterns. As a result, the model became better at distinguishing between different air-drawn digits. The accuracy improved from 8% to 54% implying that the performance was better and the model had adapted to my style.

3) Did fine-tuning help your specific gesture style?

Yes, fine-tuning clearly helped my specific gesture style. Since air-drawn digits can vary among people in terms of speed, stroke order, motion and shape, the original pretrained model didn't recognize my style of digits very well. After training on my own collected dataset, the model adapted to my style of patterns and produced more reliable predictions. The higher recall values for several digits and the improved confusion matrix confirm that the model learned my drawing style more effectively after fine-tuning.